

Spill-Detection Segmentation Project

Model Training Plan — Decision Log

1. Model & Input

Topic	Options Discussed	Final Decision	Reason
Architecture	U-Net, YOLO-seg, Mask2Former, SegFormer	SegFormer (MiT-B1, ADE20K pretrained)	Global context via transformer encoder + lightweight MLP decoder; strong pretrained checkpoints suit small dataset; real-time capable
Backbone size	B0, B1, B2–B5	B1	More capacity than B0 without high overfitting risk on ~1000 images; B2+ unnecessary and slower
Input resolution	512x512, 640x640, 960x960	640x640	960 gives marginal accuracy gain but risks batch size/time under T4's 4-hour session limit
Resize strategy	Direct resize (distorts shape), Letterbox (pad, preserve shape)	Letterbox resize	72% of dataset is non-square; direct resize would distort spill shape and hurt edge accuracy

2. Batch & Precision

Topic	Options Discussed	Final Decision	Reason
Precision	FP32, Mixed precision (AMP)	AMP (fp16)	Halves memory use, speeds up training on T4 Tensor Cores, negligible accuracy cost
Real batch size	8, 16 (memory dependent)	8	Safe, conservative fit for T4 at 640x640 with SegFormer-B1, avoids OOM risk
Gradient accumulation	x1, x2, x4	x2 (effective batch 16)	Gives stable effective batch size for gradient estimates without extra memory cost

3. Data Augmentation

Topic	Options Discussed	Final Decision	Reason
Flip	Horizontal, Vertical	Horizontal only	Vertical flip creates physically implausible floor/lighting scenes
Rotation	None, +/-10-15 degrees	+/-10-15 degrees	Cameras are rarely perfectly level in real deployments
Scale/Crop	Random crop+resize, Scale-jitter+resize	Scale-jitter (0.8x-1.2x) + resize	Random crop risked removing small spills from frame entirely; scale-jitter avoids this failure mode

Topic	Options Discussed	Final Decision	Reason
Photometric	Brightness/contrast/gamma jitter	Included	Indoor/outdoor mix has variable lighting; improves generalization across camera types
Glare/Reflection	Skip, Simulated glare augmentation	Included	Directly targets known false-positive risk: shiny/reflective floor mistaken for spill

4. Loss Function & Class Imbalance

Topic	Options Discussed	Final Decision	Reason
Loss function	CE alone, Dice alone, CE+Dice, Tversky	CE + Dice (equal weight)	Dice handles foreground/background pixel imbalance; equal weighting matches the goal of balanced FP/FN (Tversky's asymmetry not needed)
Pixel-level class weighting	Add CE class weights, or rely on Dice	No extra CE weighting	Dice already addresses imbalance; adding weighting on top risks over-predicting spill regions (more false positives)
Negative (no-spill) images	Keep at 60, increase to ~120-150	150 total (~12.6% of dataset)	Brings negatives into standard 10-15% range for false-positive suppression without diluting positive signal
Negative sampling	Plain proportional split, Weighted oversampling	Plain proportional split (no weighted sampler)	Sufficient once negatives reach ~12.6%; weighted sampler kept as a later lever if false positives persist

5. Dataset Split Strategy

Topic	Options Discussed	Final Decision	Reason
Split ratio	70/15/15, 80/10/10	80/10/10	Maximizes training data for a small (~1100 image) dataset while keeping val/test usable
Leakage prevention	Random shuffle, Group-based split, K-fold	pHash near-duplicate grouping + group-level split	Random shuffling does not prevent leakage; grouping near-duplicate/same-scene images avoids train/val contamination
Final verification	Fully automated, Fully manual, Combined	Manual eyeball check on val/test only	Val/test (~110-165 images) is small enough to scan quickly as a final sanity check after automated grouping

6. Optimizer, Learning Rate & Scheduler

Topic	Options Discussed	Final Decision	Reason
Optimizer	AdamW, Adam, SGD	AdamW	Standard choice for transformer backbones; matches original SegFormer training recipe
Learning rate setup	Single LR for all params, Differential LR (encoder vs decoder)	Differential LR: encoder 6e-5, decoder 6e-4	Protects pretrained encoder features from being overwritten while letting randomly-initialized decoder head converge faster
Weight decay	0, 0.01, 0.05	0.01	Matches AdamW/SegFormer default recipe
LR schedule	Cosine decay, Polynomial decay, Step decay	Linear warmup (~5% steps) + polynomial decay (power=1) to 1e-6	Matches original SegFormer recipe; warmup prevents early instability from randomly-initialized decoder head

7. Epochs & Checkpointing

Topic	Options Discussed	Final Decision	Reason
Max epochs	40, 60-80, 100+	80 (soft cap)	Sufficient for pretrained transfer learning on ~1000 images without excessive overfitting risk
Early stopping metric	Val loss, Val mIoU	Val mIoU	Directly reflects segmentation quality; val loss can plateau while actual mask quality still changes
Early stopping patience	5, 10, 15 epochs	10 epochs	Tolerates noisy fluctuation on a small (~110 image) val set without stopping too early
Checkpoint frequency	Every 200-300 steps, Every epoch	Every epoch (overwrite latest + save best)	Epochs run ~1-2 minutes; per-epoch checkpointing is simpler with negligible risk given short epoch time
Checkpoint storage	Local Colab disk only, Google Drive	Local save then copy to Google Drive	Local Colab storage is lost on disconnect; Drive persists across sessions given the 4-hour limit
Final model selection	Last epoch, Best val mIoU checkpoint	Best val mIoU checkpoint	Protects against late-stage overfitting

8. Evaluation Metrics

Topic	Options Discussed	Final Decision	Reason
Primary metric	Val loss, mIoU, Dice score	Val mIoU	Standard, interpretable segmentation quality measure; drives early stopping

Topic	Options Discussed	Final Decision	Reason
Additional tracking	Overall mIoU only, + Precision/Recall, + Per-size-bucket IoU	Precision/Recall + per-size-bucket IoU (diagnostic, not used for early stopping)	Overall mIoU can hide small-spill failures since large spills dominate pixel counts; per-bucket tracking directly measures small vs big spill accuracy